
radar window
60 frames, 2 s
(x, y, z, v, ι)
ROI + moving-point centring + 1.2 m gating
G=12 segments
$\times P$=32 pts
$\mathbf{X} \in \mathbb{R}^{12\times32\times6}$
Segment tokenizer
shared point MLP 6→64
point self-attn. + rel.-pos. bias
attentive & max pooling
Motion descriptor
centroid, Δcentroid,
$\overline{|v|}$, $\log n \rightarrow$ MLP
Temporal encoder
[CLS] + 12 tokens
2× Transformer block
(d=64, 4 heads)
LN + linear
$\hat{y}$
KD:
$\mathrm{KL}(p_T^\tau \| p_S^\tau)$
STPT-Large teacher
(d=128, 4 blocks)